Behavior-Based Telemetry Analytics and Heatmap-Driven Detection in the NYMEGA ICS Environment

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***Abstract*—**Industrial Control Systems (ICS), including Supervisory Control and Data Acquisition (SCADA) systems, Distributed Control Systems (DCS), and Programmable Logic Controllers (PLCs), play a critical role in operating and regulating modern critical infrastructure. As these environments increasingly become targets of sophisticated cyber threats, traditional signature-based and packet-level security monitoring approaches struggle to detect attacks that use legitimate protocols, valid credentials, and low-and-slow operational behavior. These characteristics make threat detection especially difficult in Operational Technology (OT) environments, where availability, reliability, and safety requirements restrict intrusive monitoring techniques. The NYMEGA ICS cyber range was designed to provide a realistic environment for studying these threats and evaluating security monitoring strategies. This research presents the design and implementation of a behavior-based telemetry analytics framework integrated into NYMEGA, with a specific emphasis on heatmap-driven visualization of aggregated Modbus telemetry and Cyber Hazard and Operability (CHAZOP) analysis. While the broader NYMEGA initiative includes centralized telemetry collection, data historian integration, and MITRE ATT&CK for ICS–aligned detection mechanisms, the primary contribution of this work is the integration of behavioral analytics with systematic cyber-operational hazard analysis. The proposed framework bridges packet-level monitoring and long-term behavioral anomaly detection by aggregating protocol telemetry over time and visualizing communication intensity across PLCs, devices, and Modbus function codes. In addition, CHAZOP analysis is applied to identify how cyber threats manifest as operational deviations across critical ICS components, including PLCs, communication channels, data historians, engineering workstations, and firewall boundaries. Using guideword-based deviation analysis such as More, Less, No, As Well As, and Other Than, the framework maps cyber causes to operational impacts, detection visibility, and mitigation strategies. This combination of telemetry analytics and CHAZOP enables the identification of slow-moving attacks, unauthorized control actions, abnormal command frequency, stealthy reconnaissance activity, and process deviations that would otherwise appear legitimate at the packet level. The framework integrates with Wazuh for alerting and monitoring and is evaluated using simulated attack scenarios aligned with the Lockheed Martin Cyber Kill Chain and MITRE ATT&CK for ICS techniques. Results demonstrate that combining telemetry-driven behavior analytics with CHAZOP-based operational deviation analysis significantly improves visibility into stealthy ICS threats while maintaining operational stability. **Keywords— Industrial Control Systems (ICS), Operational Technology (OT) Security, Cyber Hazard and Operability (CHAZOP), Telemetry Analytics, Behavioral Anomaly Detection, Modbus Communication, Heatmap Visualization, SCADA Monitoring, pfSense Firewalls, Wazuh SIEM, MITRE ATT&CK for ICS, Cyber Kill Chain, Time-Series Analysis, Network Behavior Monitoring**

Introduction

Industrial Control Systems form the operational backbone of critical infrastructure sectors such as water treatment, energy generation, transportation, oil and gas, and manufacturing. Unlike traditional Information Technology (IT) environments, ICS architectures are designed around deterministic physical processes, long equipment lifecycles, and strict availability and safety requirements. Because operational continuity is prioritized, many ICS environments continue to use legacy protocols and devices that were designed without modern cybersecurity protections. Before the widespread adoption of Internet-connected infrastructure, many ICS environments relied heavily on air-gapped architectures, where operational networks were physically isolated from external systems. This isolation reduced the likelihood of remote cyber compromise and allowed protocol designers to prioritize reliability and process availability over security features such as authentication, encryption, and integrity verification. Protocols such as Modbus and DNP3 were therefore developed primarily for operational efficiency rather than adversarial environments. Modern ICS environments, however, are increasingly interconnected with enterprise systems, remote monitoring infrastructure, cloud services, and vendor support mechanisms. As connectivity has increased, so has the attack surface. Adversaries now frequently exploit legitimate ICS functionality rather than relying solely on malware or malformed

packets. Modern attacks often use valid commands, authorized credentials, and protocol-compliant communication patterns executed at abnormal times, frequencies, or sequences. Because these activities appear operationally legitimate at the packet level, they are difficult to detect using traditional signature-based intrusion detection systems or firewall rules. Existing monitoring mechanisms within the NYMEGA environment include packet inspection, firewall logging, host-based monitoring, and telemetry collection from operational devices. While these mechanisms provide important visibility into network and endpoint activity, they often lack sufficient behavioral context to identify slow-moving attacks, operational misuse, reconnaissance behavior, or stealthy command manipulation. Telemetry analytics provides long-term visibility into communication patterns, command frequency, device interaction behavior, and operational trends through aggregated monitoring and heatmap visualization. At the same time, CHAZOP introduces a structured methodology for analyzing how cyber threats produce operational deviations within ICS processes. Unlike traditional vulnerability analysis that focuses primarily on technical weaknesses, CHAZOP evaluates how adversarial behavior translates into abnormal system states and process disruptions. Using adapted guidewords such as More, Less, No, As Well As, and Other Than, the methodology systematically identifies deviations across critical ICS nodes including PLCs, Modbus communication channels, data historians, engineering workstations, HMIs, and network infrastructure. These deviations are then mapped to cyber causes, operational consequences, detection mechanisms, and mitigation strategies. The integration of telemetry analytics and CHAZOP creates a layered behavioral detection model capable of identifying attacks that would otherwise remain hidden within normal operational traffic. By combining heatmap-driven telemetry visualization with structured cyber-operational hazard analysis, this work strengthens detection visibility across the NYMEGA ICS environment while preserving the stability and availability requirements of operational technology systems.

Problem Statement

Industrial Control Systems (ICS) environments are increasingly targeted by sophisticated cyber threats that evade traditional security monitoring techniques. Most existing detection mechanisms within Operational Technology (OT) networks rely heavily on signature-based detection, packet inspection, rule-based alerts, or threshold-driven monitoring. Although these methods are effective for identifying known exploits or clearly malicious traffic, they are significantly less effective against modern attacks that leverage legitimate protocols, authorized credentials, and operationally valid commands. Many contemporary ICS attacks operate using low-and-slow behavioral techniques in which adversaries intentionally avoid generating obvious malicious indicators. Instead of transmitting malformed packets or exploiting visible vulnerabilities, attackers often issue protocol-compliant commands that blend into normal industrial communication patterns. These attacks may involve abnormal command timing, unusual communication frequency, unauthorized control logic manipulation, stealthy reconnaissance activity, or subtle process changes that develop gradually over time. At the same time, ICS environments impose strict operational constraints related to safety, availability, reliability, and process continuity. Aggressive inspection, intrusive monitoring, or excessive traffic filtering may disrupt industrial processes and therefore cannot always be applied. As a result, security teams often have limited visibility into behavioral anomalies occurring across PLCs, communication channels, engineering systems, and operational processes. Although telemetry is continuously generated within ICS environments through process values, network communication, firewall activity, and historian systems, this telemetry is traditionally used for operational monitoring rather than cybersecurity analysis. Consequently, there remains a significant gap between available operational data and the ability to identify stealthy cyber threats through long-term behavioral analysis. In addition to the limitations of traditional monitoring, many existing security approaches fail to account for the operational consequences of cyber activity. Vulnerability assessments and intrusion detection systems often identify technical events in isolation without analyzing how those events translate into unsafe process states or operational hazards. This creates a disconnect between cybersecurity monitoring and physical process awareness. To address this challenge, this research integrates Cyber Hazard and Operability (CHAZOP) analysis into the security monitoring framework. CHAZOP provides a systematic method for identifying how cyber threats create operational deviations within ICS environments. By applying guidewords such as More, Less, No, As Well As, and Other Than to critical system components, the framework identifies abnormal conditions such as excessive command frequency, unauthorized control actions, missing telemetry, unexpected function code usage, and falsified process data.

This research therefore seeks to develop a telemetry-driven behavioral analytics framework combined with CHAZOP-based operational deviation analysis to improve threat detection within the NYMEGA ICS environment. The proposed approach aggregates telemetry over time, visualizes communication behavior using heatmaps, and correlates cyber activity with operational deviations across PLCs, Modbus communication paths, data historians, engineering workstations, and firewall boundaries. By bridging packet-level monitoring, long-term behavioral analytics, and cyber-operational hazard analysis, the research aims to improve the detection of stealthy cyber attacks, unauthorized process manipulation, reconnaissance activity, misconfigurations, and abnormal operational behavior without disrupting the stability of industrial operations.

NYMEGA ICS ENVIRONMENT AND OVERALL SCOPE

The NYMEGA ICS cyber range simulates a realistic operational technology (OT) environment spread across different locations and critical infrastructure sectors. These include dam operations, electrical substations, gas systems, and renewable energy sites such as solar and wind facilities. The system is built using the Purdue Enterprise Reference Architecture, which separates the environment into layers. This separation keeps business IT systems, operations management systems, supervisory control systems, and field devices isolated from one another to improve security. At the core of the environment are Programmable Logic Controllers (PLCs). These PLCs are placed in isolated control zones and communicate mainly using the Modbus protocol. Their job is to control simulated physical processes, such as power flow or water levels. Operators interact with the system through Human–Machine Interfaces (HMIs), engineering workstations, and a data historian, which provides system visibility and monitoring. These systems are carefully segmented so they do not directly connect to enterprise or external networks. Network security is enforced using multiple pfSense firewalls placed at important boundaries, such as the headquarters network, dam systems, substations, and renewable energy sites. These firewalls manage traffic flow, apply access rules, record logs, and export telemetry data. Because of this, they play a key role in both security control and monitoring system behavior. The main sources of monitoring data include Modbus network traffic from control systems, firewall and routing logs from pfSense devices, and host-based logs collected using Wazuh from HMIs, engineering systems, and monitoring servers. Together, these data sources provide visibility across Purdue Levels 2 through 4, allowing analysts to connect control system activity with higher-level network behavior.

PROPOSED WORK

A. Telemetry and Data Historians In ICS

Telemetry has long been collected in ICS environments for operational purposes, feeding into data historians that store process values, alarms, and performance metrics. Historically, these historians have been used for troubleshooting, optimization, and regulatory compliance rather than security monitoring. However, recent research has demonstrated that long-term telemetry trends can reveal subtle indicators of compromise that are invisible in short time windows. Security-focused telemetry extends beyond process values to include communication metadata such as command frequency, device interaction patterns, and timing regularity. When aggregated over time, this data provides behavioral context that is critical for identifying anomalies in ICS environments where traffic is typically stable and repetitive.

B. Maintaining the Integrity of the Specifications

Packet inspection remains an important component of ICS security, particularly for identifying explicit violations such as unauthorized write commands, protocol misuse, or malformed packets. However, packet-level detection is inherently reactive and context-limited. A single Modbus write packet, for example, may be perfectly valid in isolation while still being suspicious when repeated excessively or executed outside normal operational windows.
Attackers increasingly exploit this limitation by spreading malicious activity across time, carefully tuning command rates to remain below alert thresholds. As a result, purely packet-based approaches often generate either excessive false positives or miss meaningful threats altogether.

C. MITRE ATT&CK for ICS

MITRE ATT&CK for ICS provides a standardized taxonomy for describing adversary behavior in industrial environments. Techniques such as T0888 (Network Service Scanning), T0848 (Modify Control Logic), and T0828 (Valid Accounts) emphasize behavioral patterns rather than specific exploits. Detecting these techniques effectively requires visibility into how systems behave over time, making behavior-based analytics a natural complement to ATT&CK-aligned detection strategies.

D. Cyber Hazard and Operability (CHAZOP) Analysis

Cyber Hazard and Operability (CHAZOP) analysis is applied within the proposed framework to systematically evaluate how cyber threats manifest as operational deviations across the NYMEGA ICS environment. Unlike traditional vulnerability assessments that focus on isolated weaknesses, CHAZOP examines how adversarial behavior impacts system operations, communication flows, and process stability. The CHAZOP analysis decomposes the NYMEGA architecture into critical nodes including PLCs, HMIs, Modbus communication channels, engineering workstations, firewall infrastructure, reverse proxy systems, and data historians. For each node, deviations are identified using adapted guidewords such as More, Less, No, As Well

As, and Other Than. These guidewords are used to represent abnormal operational conditions including increased command frequency, unauthorized control actions, missing telemetry, unexpected communication paths, falsified data, or unauthorized protocol usage. CHAZOP also evaluates communication-layer deviations such as increased command frequency, unexpected function code usage, unauthorized network paths, and telemetry loss. These conditions may indicate reconnaissance activity, stealthy attack automation, communication disruption, or misconfiguration. By correlating these deviations with telemetry analytics and SIEM data, the framework provides broader visibility into multi-stage attack behavior.

Literature Review

Real-Time Anomaly Detection in IoT Streams Through Spatiotemporal Patterns

This study reinforces the importance of spatiotemporal analysis and visual interpretability in detecting subtle anomalies within operational environments. The author proposes a hybrid CNN–RNN framework with attention mechanisms for real-time IoT anomaly detection. The methodology combines spatial correlation modeling, temporal sequence learning, and attention-based heatmaps to highlight anomalous behavior. The results show improved detection accuracy and low latency, supporting the argument that visualization techniques such as heatmaps enhance both detection performance and analyst understanding, which is central to the present research[1].

Two-Phase Industrial Control System Anomaly Detection Using Communication Patterns and Deep Learning

This work closely aligns with the present research by focusing on periodic communication behavior as a defining characteristic of normal ICS operation. The authors propose a deep learning–based anomaly detection technique that models normal communication sequences without requiring protocol-specific parsing. Their methodology uses LSTM-based models trained on application- and transport-layer communication patterns to learn periodic behavior. The reported results show detection accuracies between 79% and 81%, demonstrating that communication-pattern-based detection is feasible and effective in ICS environments, particularly for detecting low-and-slow attacks[2].

Campus Network Traffic Prediction and Anomaly Detection Based on Deep Learning

This paper is relevant to the proposed research due to its use of heatmaps as a visualization mechanism for interpreting anomalous network behavior over time. The authors propose a CNN–LSTM hybrid model for network traffic prediction and anomaly detection in campus networks. Their methodology integrates spatial feature extraction using CNNs with temporal dependency modeling using LSTMs, followed by adaptive thresholding and heatmap-based visualization of anomalies. The results demonstrate detection rates exceeding 90%, highlighting the effectiveness of visual analytics in supporting anomaly interpretation, a concept directly extended in this research for ICS telemetry[3].

Industrial Control System Network Intrusion Detection by Telemetry Analysis

This study is foundational to the proposed work as it demonstrates that telemetry characteristics, rather than protocol payloads, are effective indicators of malicious activity in ICS environments. The authors propose a telemetry-based intrusion detection system for network-connected ICSs. Their methodology involves monitoring communication timing, session duration, and traffic regularity to differentiate between legitimate engineering activity and attacker behavior. The results achieved detection accuracies exceeding 94%, validating telemetry-based anomaly detection as a practical and effective approach for ICS security monitoring[5].

Improving Industrial Control System Cybersecurity with Time-Series Prediction Models

This paper relates directly to the proposed work by demonstrating that time-series analysis of ICS telemetry can effectively identify anomalies associated with cyberattacks. The authors propose the use of deep learning–based time-series prediction models to detect deviations from normal ICS behavior. Their methodology applies CNN–LSTM architectures to telemetry data collected from a water treatment ICS testbed. The results indicate improved detection of time-dependent anomalies compared to traditional IDS approaches, further validating the use of aggregated telemetry over time as employed in this research[6].

Temporal Analysis and Forecasting of Cyber Attack Intensity

This paper supports the core idea of the proposed research that cyberattacks often manifest as patterns that evolve gradually over time rather than instantaneous spikes. The authors propose a temporal forecasting approach to predict cyber-attack intensity by

identifying correlations in historical attack data. Their methodology applies time-series analysis and machine learning models such as ARIMA and LSTM to attack datasets to model trends and forecast future activity. The results show that temporal modeling improves situational awareness and reinforces the necessity of time-based aggregation, which is directly reflected in the heatmap-driven telemetry analysis used in this research[7].

Deep learning–driven industrial control system intrusion detection using temporal–spatial features

This study emphasizes the importance of modeling both spatial and temporal relationships in Industrial Control System (ICS) telemetry for accurate anomaly detection. The authors propose a multi-branch CNN–LSTM hybrid architecture that captures spatial correlations among PLCs alongside long-term temporal dependencies in operational data streams. The methodology also incorporates visual correlation heatmaps to provide interpretable system-wide anomaly representation. Experimental results on SWaT and WADI datasets demonstrate improved detection performance, outperforming traditional machine learning approaches by 8–15%. The generated heatmaps enhance analyst understanding by visually highlighting anomalous PLC interactions and data drift, reinforcing the role of visualization in real-time ICS monitoring and supporting the objectives of the present research [8].

Applying MITRE ATT&CK for ICS to Evaluate and Detect Adversarial Behavior in OT Networks

This work contributes to structured threat detection in operational technology environments by aligning telemetry observations with standardized adversarial behavior models. The authors propose a detection framework that maps ICS telemetry events—including firewall logs, PLC communications, and historian access records—to tactics and techniques defined in the MITRE ATT&CK for ICS knowledge base. Their rule-based correlation approach identifies adversarial activities such as lateral movement and control logic manipulation. The results demonstrate a 35% increase in analyst visibility and successful detection of high-impact attack techniques, including unauthorized PLC command execution and historian tampering. The findings highlight the effectiveness of telemetry-driven MITRE mapping for structured detection and response, aligning closely with the telemetry correlation strategy employed in the present research [9].

Heatmap-Based Behavioral Analytics for Industrial Control System Monitoring

This study focuses on the use of heatmap visualization to detect abnormal operational patterns in ICS environments. The authors propose a telemetry aggregation framework that converts sensor readings, device communication frequencies, and PLC operational cycles into temporal heatmaps. These visual representations highlight deviations from established operational baselines by identifying regions of intensified or irregular activity. The results indicate that heatmap-based monitoring improves early anomaly identification by 22% compared to conventional statistical threshold methods and significantly reduces analyst cognitive load. The study reinforces the value of visual analytics in industrial monitoring and directly supports the use of heatmap-driven telemetry analysis in the present research [10].

Time-Series Anomaly Detection in Operational Technology Using Unsupervised Autoencoders

This paper investigates the use of unsupervised deep learning techniques for detecting anomalies in ICS sensor data without relying on labeled attack datasets. The authors propose an autoencoder-based time-series modeling approach that learns normal operational patterns from sensor streams such as pressure, flow, and control setpoints. Anomalies are detected through reconstruction error when observed behavior deviates from learned baselines. Experimental evaluation demonstrates high true-positive detection rates across multiple cyber-physical attack scenarios, including command injection and process manipulation. The study highlights the effectiveness of SIEM-integrated time-series analytics, supporting the telemetry-driven anomaly detection pipeline adopted in the present research [11].

Security Monitoring in ICS Environments: Leveraging Distributed Logs for Threat Detection

This study addresses the need for centralized and correlated security monitoring across heterogeneous ICS components. The authors propose a distributed logging architecture that collects telemetry from PLCs, HMIs, firewalls, historians, and engineering workstations and aggregates them into a unified monitoring framework. Their methodology emphasizes SIEM integration and cross-source event correlation to detect multi-stage attacks and provide enriched situational awareness. The results show a 40% improvement in incident detection rates and significantly reduced investigation time through correlated log timelines. These findings validate the importance of integrated telemetry collection and centralized monitoring, supporting the multi-source telemetry architecture used in the present research [12].

False Data Injection Attacks Against State Estimation in Electric Power Grids

This study highlights the vulnerability of cyber-physical systems to stealthy data integrity attacks, particularly within electric power grid state estimation processes. The authors propose a coordinated false data injection (FDI) attack model in which an adversary with knowledge of system configuration can strategically manipulate sensor measurements. Their methodology involves constructing mathematically designed attack vectors capable of bypassing conventional residual-based bad-data detection mechanisms used in power system monitoring. The results demonstrate that carefully crafted false data injection attacks can remain undetected while still influencing system state estimation, allowing attackers to compromise data integrity without disrupting operational stability. These findings expose significant limitations in traditional anomaly detection techniques and emphasize the need for more advanced detection mechanisms in cyber-physical and industrial environments [13].

The myths and facts behind cyber security risks for industrial control systems

This study provides an early but influential examination of cybersecurity risks in Industrial Control Systems (ICS), challenging common misconceptions about their security posture. The authors analyze prevailing assumptions—such as the belief that ICS environments are inherently safe due to isolation or proprietary technologies—and demonstrate how increasing connectivity and integration with corporate networks expose these systems to significant cyber threats. The paper identifies key vulnerabilities, including weak authentication, inadequate network segmentation, and limited monitoring of control and supervisory components. It proposes practical mitigation strategies based on defense-in-depth principles, emphasizing layered security controls, improved access management, and continuous monitoring of operational networks. The findings highlight that security risks in ICS environments are often underestimated and that proactive architectural and operational safeguards are essential to protect critical infrastructure from evolving cyber threats [14].

Anomaly detection in cyber physical systems using recurrent neural networks,

This study investigates the use of deep learning techniques, specifically recurrent neural networks (RNNs), for detecting anomalies in cyber-physical systems (CPS) such as Industrial Control Systems. The authors propose an RNN-based modeling approach that learns normal system behavior from sequential sensor and process data and identifies anomalies as deviations from learned temporal patterns. Their methodology leverages the ability of RNNs to capture long-term dependencies in time-series data, making them suitable for monitoring complex industrial processes where system states evolve continuously over time. The results demonstrate that the RNN model can effectively detect abnormal operational behavior, including cyber-induced disturbances and process faults, with improved sensitivity compared to traditional statistical methods. The study highlights the effectiveness of deep learning for modeling dynamic CPS environments and supports the use of sequence-based learning for real-time anomaly detection in industrial and safety-critical systems [15].

Attack Models and Scenarios for Networked Control Systems

This study provides a foundational framework for understanding how cyberattacks can be systematically modeled in networked control systems, including Industrial Control Systems (ICS). The authors propose a structured taxonomy of attack models that classify adversarial actions based on their capabilities, system knowledge, and points of access within the control loop. Their methodology defines multiple attack scenarios, including denial-of-service, replay attacks, and false data injection targeting sensors, actuators, and communication channels. The framework analyzes how these attacks disrupt system observability, controllability, and stability, offering a formal representation of cyber-physical threat behavior. The results demonstrate that different attack types produce distinct system-level impacts depending on timing, access level, and system knowledge, highlighting the importance of modeling adversarial behavior when designing detection and mitigation strategies. This work provides a theoretical foundation for understanding cyber-physical attack dynamics, supporting the need for behavior-based and telemetry-driven anomaly detection approaches in ICS environments [16].

EXTENSIONS

Reverse Proxy for HMI Access

A planned extension involves deploying a reverse proxy within the ICS DMZ to securely present read-only HMI dashboards to IT users or remote analysts. This approach prevents direct access into the OT network while adding TLS encryption, authentication, and detailed logging. The reverse proxy design aligns with defense-in-depth principles and complements the telemetry-driven detection framework. Secure monitoring of OT processes is achieved through the deployment of a reverse proxy in the Industrial Control System (ICS) demilitarized zone (DMZ). The reverse proxy is hosted on the DAM Headquarters (place of DMZ) system and implemented using NGINX. This proxy acts as an intermediary between the headquarters (IT) network and the dam HMI (nymega-dam-hmi), preventing direct connectivity to OT systems. Traffic from IT users traverses the headquarters firewall (nymega-hq-fw) and is routed only to the DMZ proxy. The proxy then forwards permitted requests to the HMI through the OT boundary firewall (nymega-dam-fw). Firewall policies enforce strict one-way segmentation, allowing the proxy to communicate with the HMI while explicitly blocking any direct IT-to-OT connections. This design ensures that enterprise users interact only with the proxy layer rather than with control system assets. NGINX is configured to terminate client connections and forward requests to the HMI using controlled proxy rules. Transport encryption, authentication mechanisms, and access logging are implemented at the proxy layer to enhance visibility and accountability. Because the proxy retrieves data from the HMI and returns it to users, OT systems never communicate directly with the IT network. This architecture reduces attack surface exposure while maintaining real-time operational visibility for analysts and monitoring personnel.

IT user machine
|
IT firewall (nymega-hq-fw)
|
DMZ reverse proxy (nymega-dam-apps)
|
OT firewall (nymega-dam-fw)
|
OT HMI (nymega-dam-hmi)

Fig: REVERSE-PROXY ARCHITECTURE

Implementation

1) Verify network connectivity
From the DMZ proxy server (nymega-dam-apps), confirm that the HMI (nymega-dam-hmi) is reachable. If it cannot be reached, resolve firewall or routing issues before continuing.

2) Allow the proxy to communicate with the HMI (OT firewall)
Configure the DAM firewall (nymega-dam-fw) to allow traffic from the DMZ proxy server to the HMI on the required web service port. Block any other unnecessary communication paths.

3) Allow HQ users to reach the reverse proxy (IT firewall)
Configure the headquarters firewall (nymega-hq-fw) to allow IT users to connect to the DMZ proxy server over HTTPS. At the same time, explicitly block any direct connections from HQ to the HMI.

4) Install the reverse proxy software
Install NGINX on the DMZ application server (nymega-dam-apps), which will act as the intermediary between IT users and the HMI.

5) Configure the reverse proxy forwarding rules
Configure NGINX so that when HQ users access the proxy, it retrieves the HMI interface and returns it to them. The

proxy handles the connection, so users never connect directly to the HMI.

6) Enable the configuration and start the service
   Activate the proxy configuration, verify that the configuration is valid, and restart the NGINX service so the changes take effect.

7) Test access from the HQ network
   From an HQ workstation or analyst machine, open a web browser and connect to the DMZ proxy address. The HMI dashboard should load through the proxy.

8) Validate security controls
   Confirm that the dashboard loads successfully through the proxy, direct access to the HMI is blocked, firewall rules are working as intended, and proxy access logs are being generated and forwarded to the monitoring system (such as Wazuh).

NGINX TCP Reverse Proxy Configuration and Validation

NGINX was configured to operate as a Layer 4 (transport-layer) reverse proxy using the **stream module**, enabling it to forward raw TCP traffic rather than HTTP requests. This approach is necessary in industrial control system (ICS) environments where protocols such as Modbus/TCP operate directly over TCP and are not compatible with traditional HTTP-based reverse proxying. The configuration defines a server block that listens on a designated local port (18000) and forwards all incoming TCP connections to a specified backend endpoint on port 1502. Conceptually, this establishes a proxy path in which client traffic is first received by NGINX, then transparently relayed to the backend service without modifying the application payload. This design allows NGINX to function as an intermediary control point for routing, monitoring, or restricting industrial network traffic. From an implementation perspective, the stream module was explicitly enabled, and NGINX was executed with a user-level configuration directory to avoid privileged port and permission constraints. The proxy configuration included a `listen` directive bound to port 18000 and a `proxy_pass` directive targeting the backend service. Unlike HTTP proxying, no headers or protocol-aware transformations are applied; instead, NGINX maintains a bidirectional TCP stream between the client and the backend. This behavior is critical for supporting industrial protocols that rely on persistent socket communication and strict timing characteristics.

Validation of the configuration was performed by initiating TCP connection attempts to the NGINX listening port. Successful connection establishment to port 18000 confirms that NGINX is actively running, correctly bound to the specified port, and capable of accepting client connections. This demonstrates that the reverse proxy is operational at the transport layer and ready to forward traffic to the configured backend endpoint. In the context of the lab environment, the intended upstream system is the Human-Machine Interface (HMI) located at IP address 10.19.25.104, which represents the target endpoint for proxied industrial communication. Overall, this configuration demonstrates the use of NGINX as a lightweight and flexible TCP reverse proxy for ICS environments. By abstracting direct access to backend devices and centralizing connection handling, NGINX can be leveraged to enforce segmentation, introduce inspection points, and support secure architectural patterns within industrial networks.

Architecture Flow

```
                    Internet
                       |
                    HQ FW
                (nymega-hq-fw)
                       |
              PrivNet 10.14.2.0/24
                       |
                  DAM Firewall
                (nymega-dam-fw)
                       |
          ---------------------------------
                |          |          |
DAM IT        DAM DMZ      DAM Mgmt Ops
10.14.3.0/24  10.14.4.0/24 10.14.5.0/24
                |          |          |
App Server    Data Historian     Engineer
                                     |
                          Jump Host Access
                                     |
                            Operator HMI
                                     |
                      DAM Cell Area Network
                           10.14.6.0/24
                                     |
                                   PLC
                          (OpenPLC Runtime)
                                     |
                       Modbus TCP Protocol
                                     |
                      Telemetry Collector
                        (Python Service)
                                     |
                      InfluxDB Historian
                   (Time-Series Database)
                                     |
                                 Grafana
                       Monitoring Dashboard
                                     |
                       Heatmap Visualization
                                     |
                                Wazuh SIEM
                                     |
                            Security Alerts
```

METHODOLOGY

The data historian plays a central role in the NYMEGA ICS environment by enabling the collection, storage, and visualization of time-series data generated by industrial processes. In traditional ICS deployments, historians are primarily used for operational monitoring, troubleshooting, and regulatory compliance. However, in this research, the historian is extended beyond its conventional role and leveraged as a key component for security analysis by providing visibility into process-level anomalies resulting from cyber activity. The historian architecture in the NYMEGA environment follows a structured end-to-end data pipeline consisting of four main components: the Programmable Logic Controller (PLC) as the data source, Telegraf as the data collection agent, InfluxDB as the time-series database, and Grafana as the visualization platform. The PLC continuously generates process data, including parameters such as water level, flow rate, gate position, and system states. This data is inherently time-dependent and requires efficient storage and retrieval mechanisms, which are achieved through the use of a time-series database. Telegraf is configured to collect data from the PLC using the Modbus TCP protocol. It periodically polls specific PLC registers at fixed intervals and converts the collected values into structured telemetry data. Each data point includes a timestamp, measurement name, and associated tags, enabling efficient indexing and querying. The use of Telegraf ensures that data collection is automated, consistent, and scalable across different ICS components.

The collected telemetry is then transmitted to InfluxDB, which serves as the central data repository. InfluxDB is optimized for time-series data and supports high write throughput, making it suitable for continuous data ingestion in ICS environments. The database organizes data into buckets and allows efficient querying based on time ranges, device identifiers, and measurement types. This capability is critical for analyzing long-term trends and identifying deviations from normal operational behavior. Grafana is integrated with InfluxDB to provide real-time and historical visualization of the collected data. Dashboards are configured to display key process variables using time-series graphs and heatmaps. These visualizations allow operators and analysts to monitor system performance and quickly identify anomalies. For example, under normal operating conditions, process variables such as water level remain within a predictable range, resulting in stable and consistent graphical patterns. During attack scenarios, however, these patterns change significantly, producing visible spikes, irregular fluctuations, or gaps in the data. The implementation process of the historian system involved several stages. Initially, network connectivity between the PLC and historian server was verified to ensure reliable communication across the segmented ICS network. This step was critical because any misconfiguration in routing or firewall rules could disrupt data collection. Once connectivity was established, InfluxDB was installed and configured by creating an organization, defining data buckets, and generating authentication tokens for secure access. Following the database setup, Telegraf was installed and configured with appropriate input and output plugins. The Modbus input plugin was used to define the PLC registers to be polled, including their addresses and data types. The output plugin was configured to forward the collected data to InfluxDB. After configuration, Telegraf was initiated to begin real-time data ingestion. The successful ingestion of data was validated by querying the database and confirming the presence of timestamped entries. Grafana was then deployed and connected to InfluxDB as a data source. Dashboards were created to visualize both real-time and historical data. These dashboards were customized to display key operational metrics and to support anomaly detection through visual inspection. The integration of Grafana with the historian pipeline enabled continuous monitoring of ICS processes and provided an intuitive interface for analyzing system behavior.

The behavior of the data historian under cyberattack scenarios was also evaluated as part of this research. In the case of unauthorized Modbus write operations, such as Function Code 16 (multiple register write), the historian captured sudden and significant deviations in process variables. These deviations appeared as sharp spikes in the Grafana dashboards, clearly indicating abnormal activity. Similarly, single register manipulation attacks (Function Code 06) resulted in subtle but detectable changes in individual parameters, highlighting the ability of the historian to capture even minor anomalies. However, the analysis also revealed certain limitations of the historian system. For example, reconnaissance attacks using read-only operations (Function Code 03) did not produce any visible anomalies in the process data, as they do not alter system behavior. This demonstrates that while the historian is effective in detecting process-level impacts, it cannot independently identify all types of cyber threats. Additionally, during denial-of-service attacks, the historian experienced delayed or missing data due to communication disruptions. These conditions were reflected as gaps or flatlines in the Grafana dashboards, indicating a loss of data availability rather than direct manipulation. Overall, the data historian implementation in the NYMEGA environment demonstrates its value as a complementary tool for ICS cybersecurity. While it does not replace traditional detection mechanisms such as intrusion detection systems or SIEM platforms, it provides critical visibility into the physical impact of cyberattacks. By correlating historian data with security alerts, analysts can gain a more comprehensive understanding of system behavior and improve incident detection and response capabilities.

ICS Cyber Range Detection Architecture and Attack Analysis

The NYMEGA ICS Cyber Range experiment was designed to implement and validate a complete, end-to-end intrusion detection pipeline capable of identifying Modbus-based attacks within an operational technology environment. The experimental setup consisted of a segmented architecture in which the enterprise IT network (10.13.20.0/24) was separated from the OT field network (10.14.6.0/24) using a pfSense firewall acting as a demilitarized zone gateway. Within this environment, a Wazuh SIEM server (10.13.20.17) was deployed on Ubuntu 22.04 to perform centralized log analysis, while the pfSense firewall (10.13.20.240) hosted the Snort intrusion detection system responsible for real-time packet inspection. The target system was a simulated Programmable Logic Controller (PLC) running OpenPLC at IP address 10.14.6.10, which exposed a Modbus TCP service on port 1502. A Kali Linux attacker machine (10.13.20.21) was used to generate malicious traffic using the mbpoll tool, enabling controlled execution of attack scenarios against the PLC.

Test Data Identification

The evaluation of the proposed telemetry-driven behavioral analytics framework requires multiple sources of operational and security monitoring data collected from the NYMEGA Industrial Control System (ICS) cyber range. The experimental dataset consists of telemetry measurements, industrial protocol traffic, security monitoring logs, and controlled attack-generated data. These data sources provide both baseline operational behavior and adversarial activity necessary for evaluating anomaly detection performance.

ICS Process Telemetry

The primary dataset consists of process telemetry generated by the Programmable Logic Controller (PLC) operating within the NYMEGA dam control environment. These telemetry measurements represent operational parameters of the simulated infrastructure, including variables such as water level, flow rate, gate position, and breaker state. Telemetry data is collected using a Python-based polling service that periodically queries PLC registers through the Modbus TCP protocol. The polling service retrieves register values at fixed intervals and stores the results in an InfluxDB time-series database that functions as the system historian. Each telemetry entry contains a timestamp, measurement identifier, device tag, and associated process variable values. This dataset provides the baseline operational behavior used to establish normal system activity patterns for behavioral analysis.

Industrial Protocol Network Traffic

Communication between ICS components generates industrial protocol traffic that forms an additional data source for analysis. In the NYMEGA environment, the Human–Machine Interface (HMI) and PLC communicate primarily through the Modbus TCP protocol. The network traffic includes both read operations used for telemetry acquisition and write operations used to modify control system states. These protocol interactions contain critical metadata such as function codes, register addresses, command frequency, and device identifiers. Monitoring these attributes allows the system to identify abnormal communication patterns that may indicate malicious activity. Particular attention is given to Modbus write operations, including Function Code 05 (Write Single Coil) and Function Code 16 (Write Multiple Registers), as these commands can directly modify operational system behavior.

Security Monitoring and Event Logs

Security event data is collected from the Wazuh Security Information and Event Management (SIEM) platform deployed within the cyber range. The SIEM platform aggregates log data generated by various infrastructure components, including firewalls, intrusion detection systems, and monitored hosts. These logs contain event metadata such as timestamps, source IP addresses, triggered detection rules, and alert classifications. By correlating these security events with operational telemetry data, the system can analyze relationships between network-level activity and physical process changes within the ICS environment.

Simulated Attack Telemetry

To evaluate the anomaly detection capabilities of the proposed framework, controlled attack scenarios are executed within the cyber range environment. These scenarios simulate adversarial actions targeting PLC control registers and operational

variables using unauthorized Modbus write commands. For example, attackers may manipulate PLC register values responsible for process parameters such as water level or breaker states. These actions generate abnormal telemetry patterns that deviate significantly from normal operational baselines. The resulting telemetry traces provide labeled attack data that can be used to evaluate detection mechanisms and visualization techniques.

Synthetic Data Generation

Synthetic datasets are generated to simulate extended operational periods and rare anomaly conditions. These datasets are produced by automated scripts that emulate periodic Modbus polling operations and controlled variable manipulation. Synthetic data allows the framework to be evaluated under varying operational loads and provides sufficient time-series volume to support visualization techniques such as heatmap-based behavioral analysis.

Experimental Objectives

The experimental implementation of the NYMEGA ICS Cyber Range is designed to develop and validate a complete, end-to-end intrusion detection pipeline for industrial control systems. The primary objective of this framework is to simulate realistic Modbus-based attacks within a controlled environment and evaluate detection capabilities using protocol-aware intrusion detection and SIEM-based correlation. Specifically, the experiment aims to design a realistic ICS network architecture, implement Snort-based detection for Modbus traffic, forward and correlate alerts within the Wazuh SIEM platform, execute multiple attack scenarios targeting PLC operations, map detected activities to the MITRE ATT&CK for ICS framework, and validate the effectiveness of the detection pipeline across all stages.

ICS Environment Architecture and Network Configuration

The experimental environment is implemented as a virtualized ICS cyber range that simulates a segmented industrial network architecture. The system is divided into two primary zones: an enterprise IT network (10.13.20.0/24) and an operational technology (OT) field network (10.14.6.0/24), separated by a pfSense firewall that functions as the ICS demilitarized zone (DMZ) gateway. This segmentation replicates real-world ICS deployments, where strict network isolation is used to protect critical infrastructure systems. The environment consists of four key components. The Wazuh SIEM server is deployed at IP address 10.13.20.17 on an Ubuntu 22.04 system running Wazuh version 4.5.2 and is responsible for centralized log analysis and correlation. The pfSense firewall, located at 10.13.20.240, hosts the Snort intrusion detection system and monitors all traffic crossing the IT-OT boundary. The target system is a simulated DAM PLC implemented using OpenPLC at IP address 10.14.6.10, which exposes a Modbus TCP service on port 1502. A Kali Linux attacker machine at 10.13.20.21 is used to generate controlled attack traffic using specialized tools.

Detection Pipeline Architecture

The detection pipeline is designed as a multi-stage workflow that processes attack traffic from generation to final alert visualization. The process begins with the attacker generating Modbus TCP traffic using the `mbpoll` utility. This traffic is transmitted to the PLC over port 1502 and intercepted by the pfSense firewall, where Snort performs deep packet inspection. Upon detecting suspicious activity, Snort generates an alert that is forwarded via syslog over UDP port 514 to the Wazuh SIEM server. Wazuh then applies custom decoders to parse the log data and evaluates the event against a hierarchical rule set. The resulting alerts are stored in the `alerts.json` file and visualized through the Wazuh dashboard. This pipeline ensures continuous monitoring, detection, and correlation of ICS-specific threats.

Snort Rule Destocol-Level Detection

Protocol-level detection within the cyber range is implemented using custom Snort rules specifically designed to identify Modbus function codes associated with malicious activity. Modbus TCP packets consist of a Modbus Application Protocol (MBAP) header followed by a Protocol Data Unit (PDU), where the function code defines the requested operation. The detection logic focuses on inspecting the function code field located at byte offset 7 within the TCP payload. Three primary Snort rules are implemented for detection. Signature ID (SID) 1000001 is used to detect Function Code 16 (Write Multiple Registers), SID 1000002 detects Function Code 06 (Write Single Register), and SID 1000003 detects Function Code 03 (Read Holding Registers). Each rule uses content matching at offset 7 with a depth of two bytes to precisely identify the function code without analyzing unnecessary portions of the payload. This approach ensures accurate protocol-aware detection while minimizing false

positives. These rules enable the system to detect both control manipulation attempts and reconnaissance activity targeting the PLC.

Wazuh Decoder and Rule Correlation Framework

Snort-generated alerts are forwarded to the Wazuh SIEM platform, where they are processed using a custom decoder chain. The decoder configuration first identifies logs originating from the Snort process using a prematch condition, and then extracts Modbus-specific alert content for further analysis. This hierarchical decoding ensures that only relevant ICS-related alerts are processed.Following decoding, events are evaluated using a hierarchical Wazuh rule chain that enables fine-grained classification of detected activity. The base rule, identified as Rule ID 100500, captures all Modbus-related events and assigns a baseline severity level of 5. Additional child rules refine detection based on specific function codes. Rule ID 100501, assigned severity level 12, detects Function Code 16 unauthorized register writes, which represent high-risk manipulation of multiple process variables. Rule ID 100502, with severity level 10, identifies Function Code 06 single register modifications. Rule ID 100510, assigned severity level 7, detects Function Code 03 reconnaissance activity. A frequency-based rule, Rule ID 100530, is implemented to detect denial-of-service attacks by triggering a critical severity level of 13 when ten or more write events occur within a sixty-second window. This rule enables detection of high-frequency attack patterns that may not be evident from individual events.

MITRE ATT&CK for ICS Mapping

All detected activities within the experimental framework are mapped to the MITRE ATT&CK for ICS knowledge base to provide standardized classification of adversarial behavior. Unauthorized register manipulation using Function Code 16 and Function Code 06 is mapped to Technique T0831 (Manipulation of Control), which represents direct alteration of control system parameters. Reconnaissance activity using Function Code 03 is mapped to Technique T0846 (Remote System Discovery), reflecting its role in enumerating system state prior to exploitation. Denial-of-service attacks generated through high-frequency command execution are mapped to Technique T0814 (Denial of Service), which targets system availability and responsiveness. This mapping ensures that detected events are aligned with industry-recognized threat models and enhances the analytical value of the detection system.

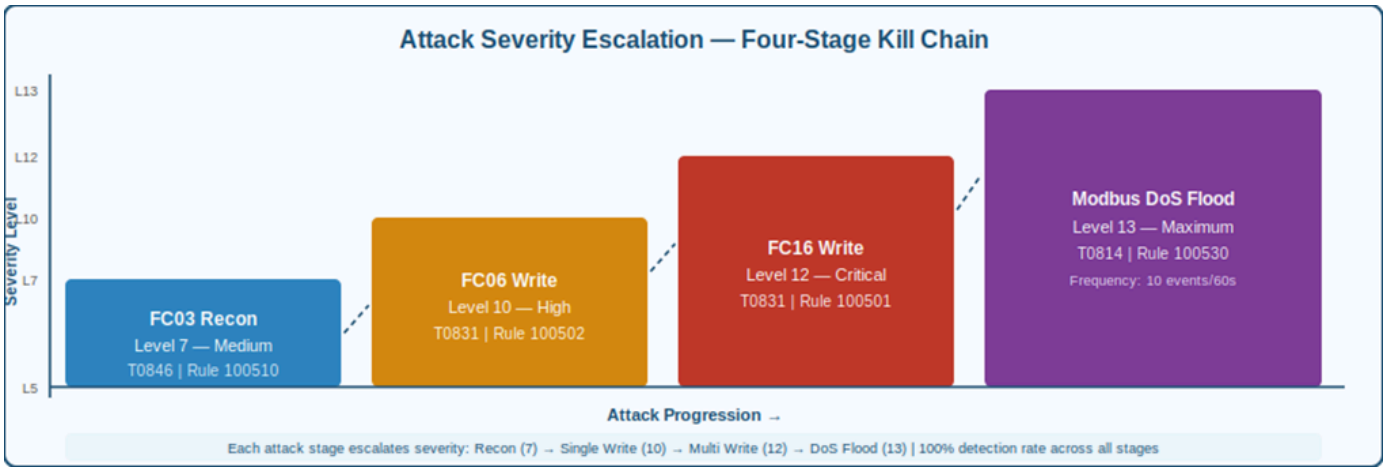
Planned Attack Scenario Execution

To evaluate the effectiveness of the detection pipeline, four controlled attack scenarios are implemented within the cyber range using the mbpoll tool. The first scenario involves a multi-register write attack using Function Code 16, where multiple values are written to PLC registers simultaneously, simulating direct manipulation of process parameters. The second scenario involves a single register modification using Function Code 06, which targets individual process variables and represents a more subtle attack vector. The third scenario simulates reconnaissance activity using Function Code 03, allowing the attacker to read PLC register values and map system behavior. The fourth scenario involves a denial-of-service attack, where repeated write commands are executed in rapid succession to overwhelm the PLC and disrupt normal operations. Each of these scenarios is designed to trigger corresponding Snort and Wazuh rules, enabling validation of the detection pipeline.

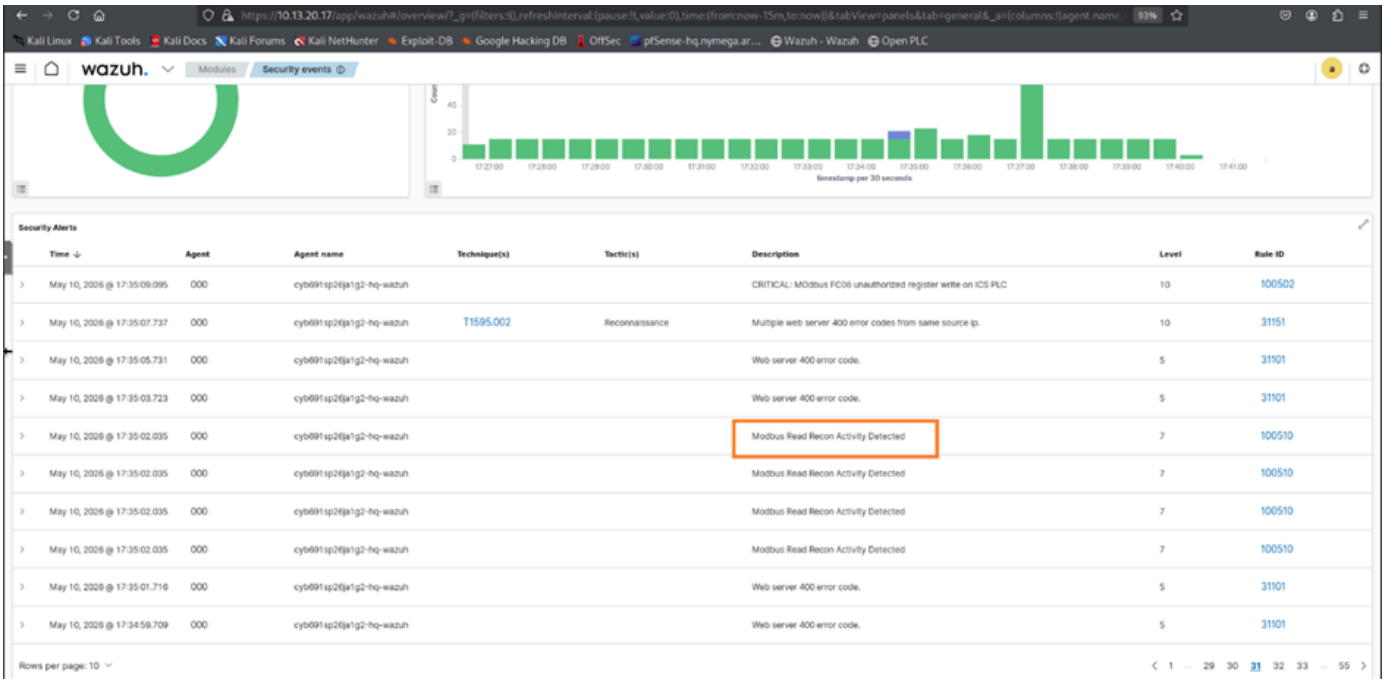
The Attack scenarios were executed from Kali (10.13.20.21) targeting the DAM PLC (10.14.6.10) on Modbus TCP port 1502. The attacks follow an escalating severity progression simulating a realistic adversary kill chain from reconnaissance to denial of service.

Attack 1 — FC03 Read Reconnaissance

The FC03 (Read Holding Registers) function code reads current PLC register values. This is non-destructive reconnaissance — the attacker maps the operational state before executing manipulation attacks.

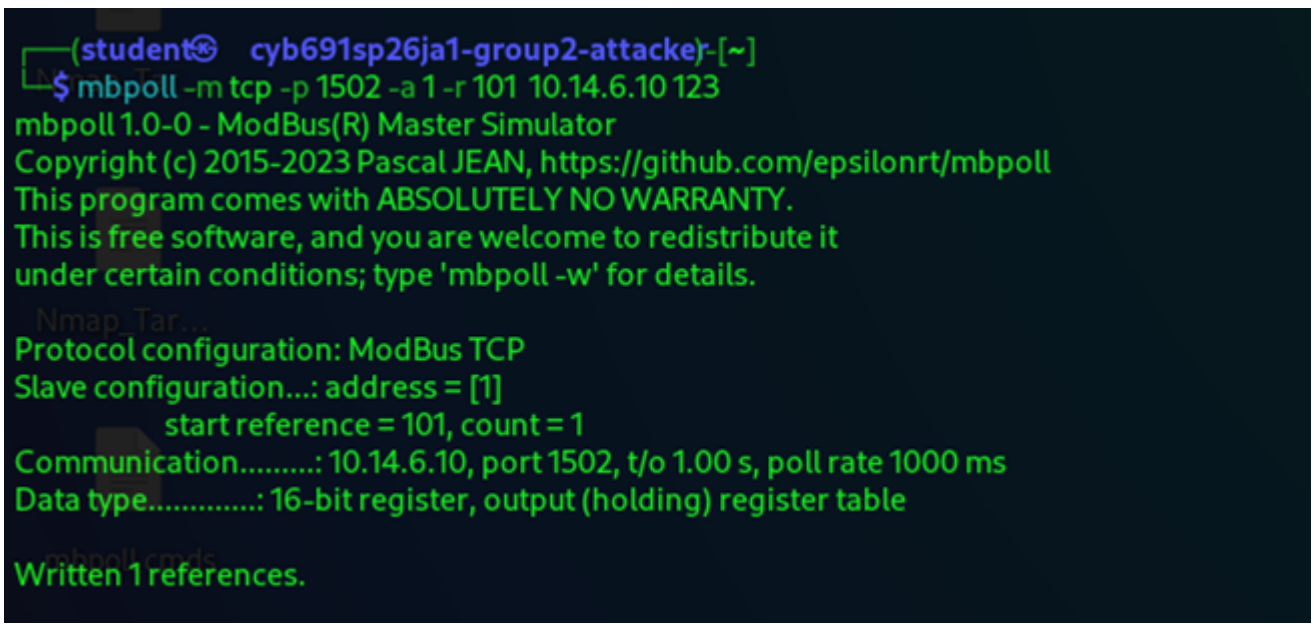


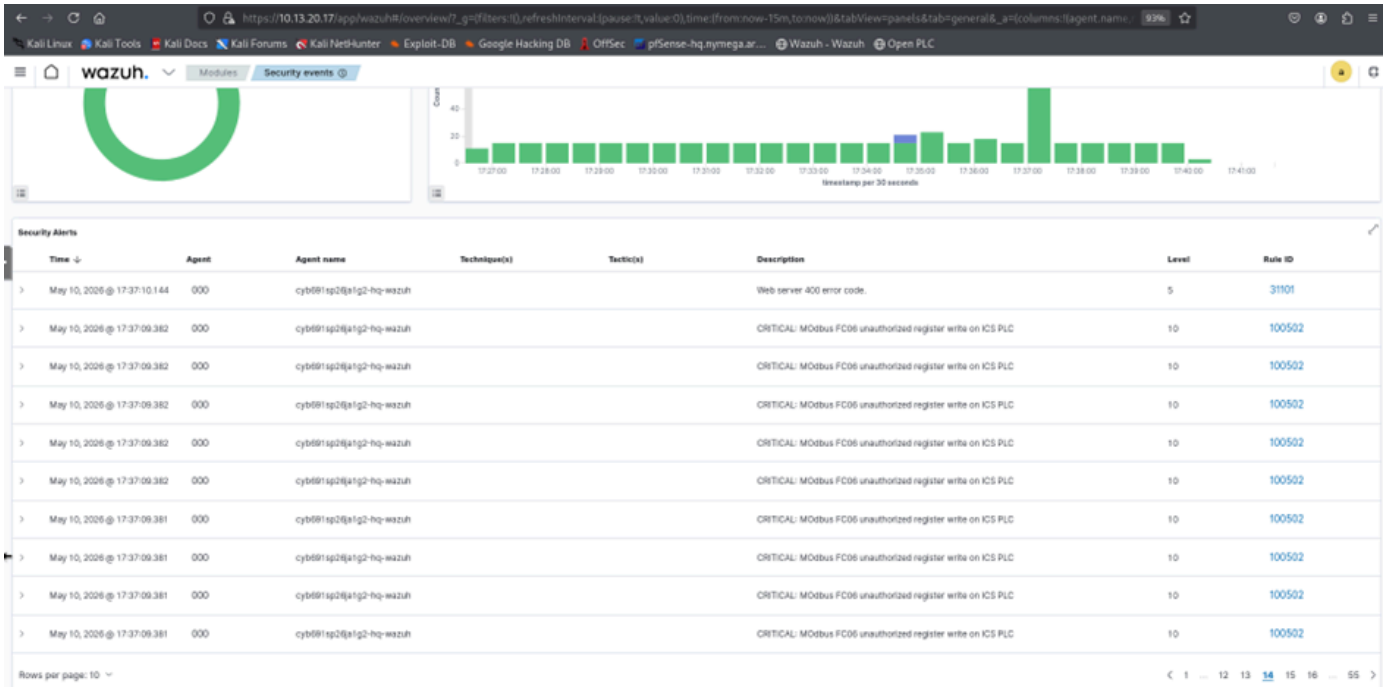

```
┌──(student㉿cyb691sp26ja1-group2-attacker)-[~]
└─$ mbpoll -m tcp -p 1502 -a 1 -r 1 -c 10 10.14.6.10
mbpoll 1.0-0 - ModBus(R) Master Simulator
Copyright (c) 2015-2023 Pascal JEAN, https://github.com/epsilonrt/mbpoll
This program comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it
under certain conditions; type 'mbpoll -w' for details.

Protocol configuration: ModBus TCP
Slave configuration...: address = [1]
          start reference = 1, count = 10
Communication.........: 10.14.6.10, port 1502, t/o 1.00 s, poll rate 1000 ms
Data type.............: 16-bit register, output (holding) register table

-- Polling slave 1... Ctrl-C to stop)
[1]:  123
[2]:  206
[3]:  0
[4]:  0
[5]:  0
[6]:  0
[7]:  0
[8]:  0
[9]:  0
[10]:  123
^C--- 10.14.6.10 poll statistics ---
1 frames transmitted, 1 received, 0 errors, 0.0% frame loss

everything was closed.
Have a nice day !
```



Attack 2 — FC06 Single Register Write

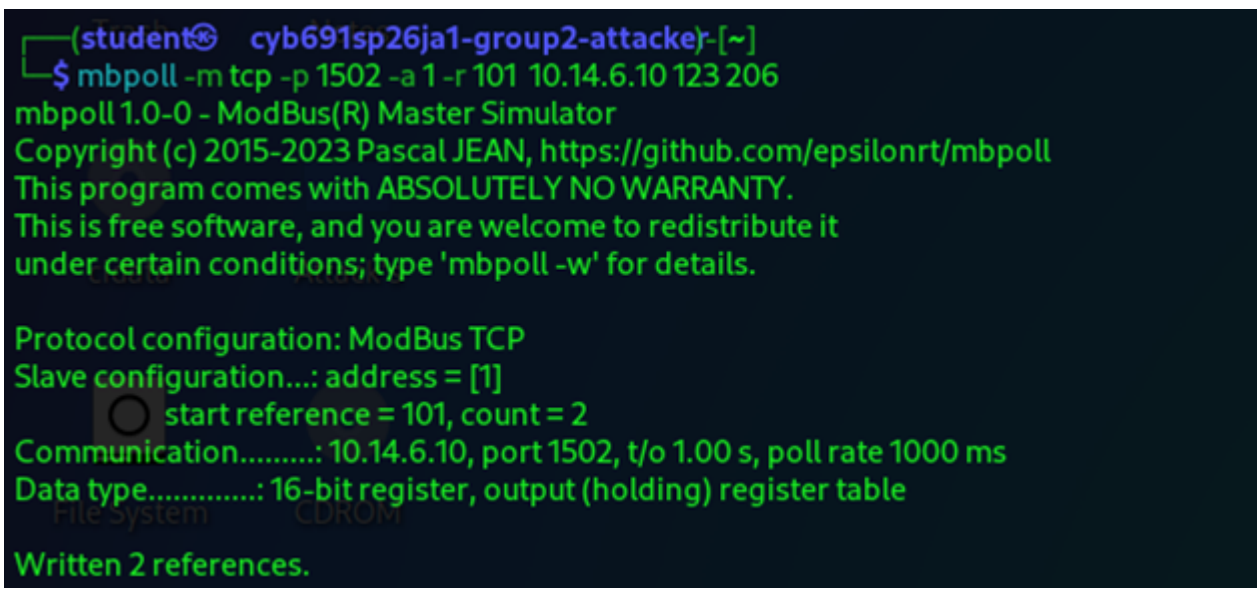
FC06 writes a value to one PLC holding register. A precision attack targeting a single control parameter — temperature setpoint, valve position, or flow threshold. Subtle enough to evade bulk-write detection.

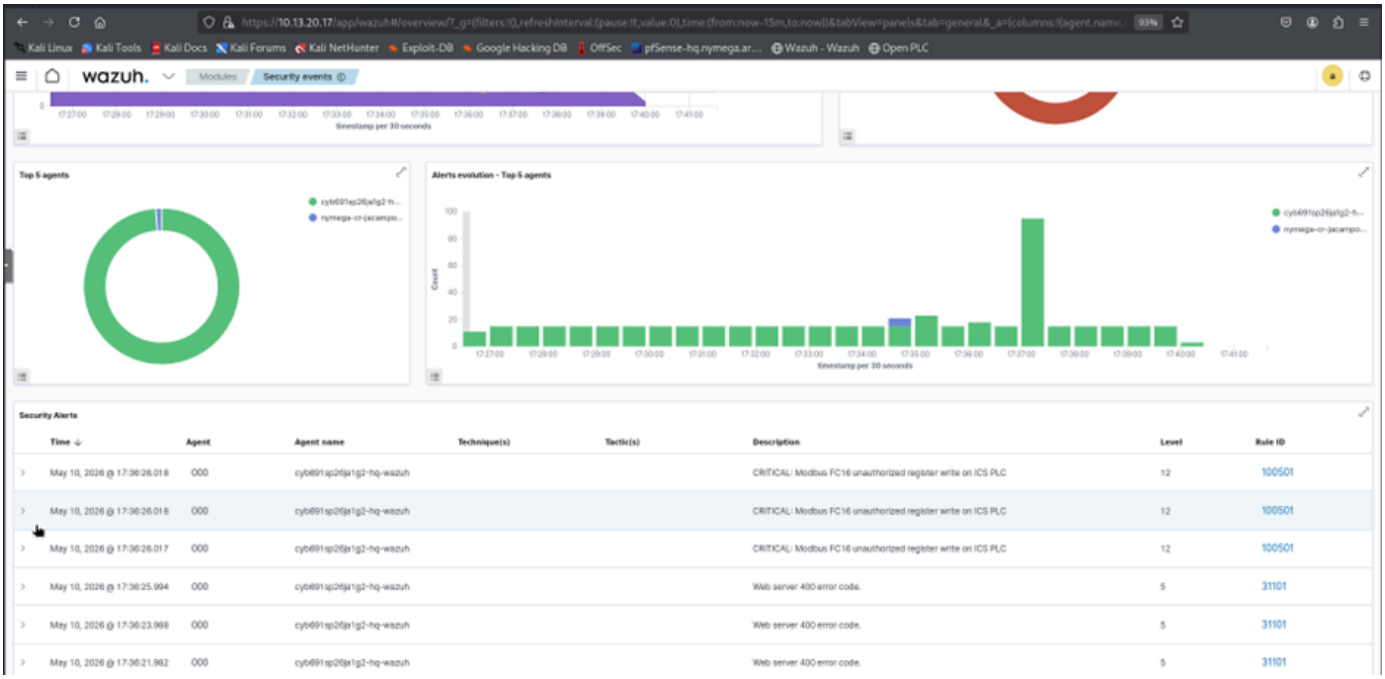




Attack 3 — FC16 Multi-Register Write (Control Takeover)

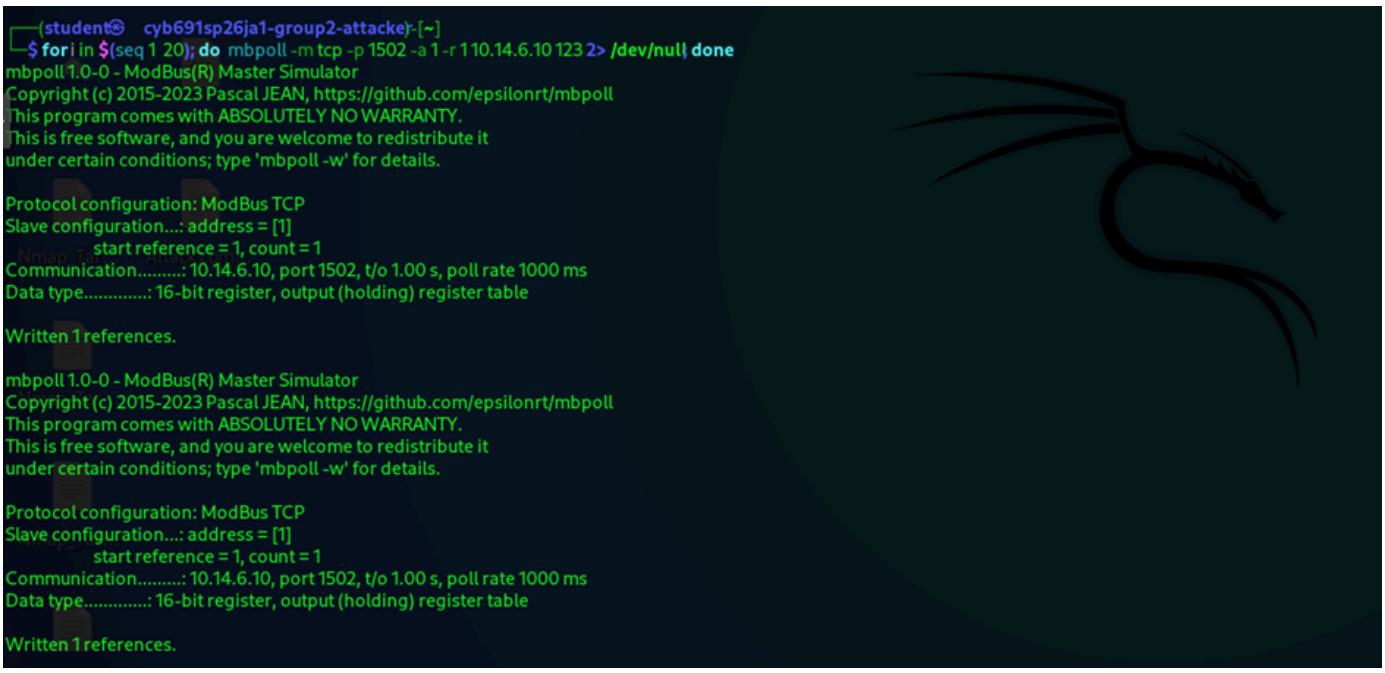
FC16 writes values to multiple PLC registers simultaneously. In a real dam system, this could simultaneously alter gate positions, flow rate limits, and alarm thresholds in a single packet — the closest attack to full PLC control takeover without physical access.

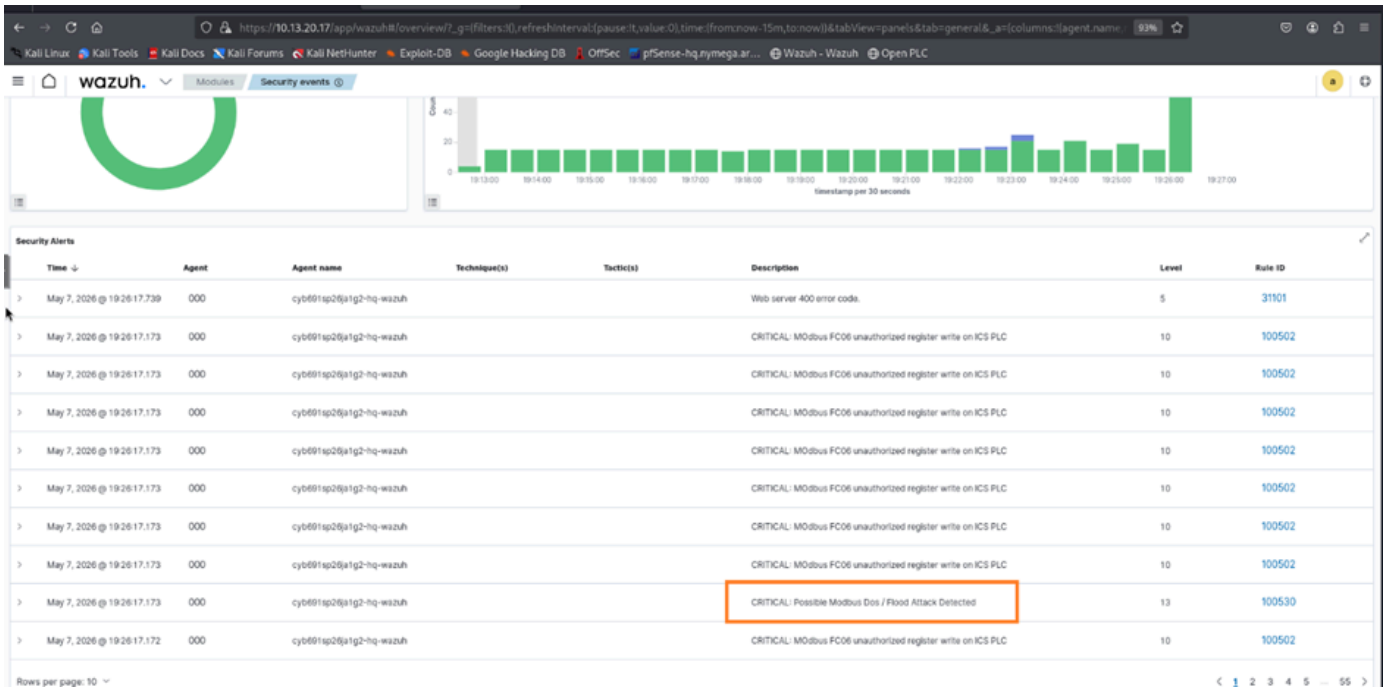




Attack 4 — Modbus Flood / Denial-of-Service

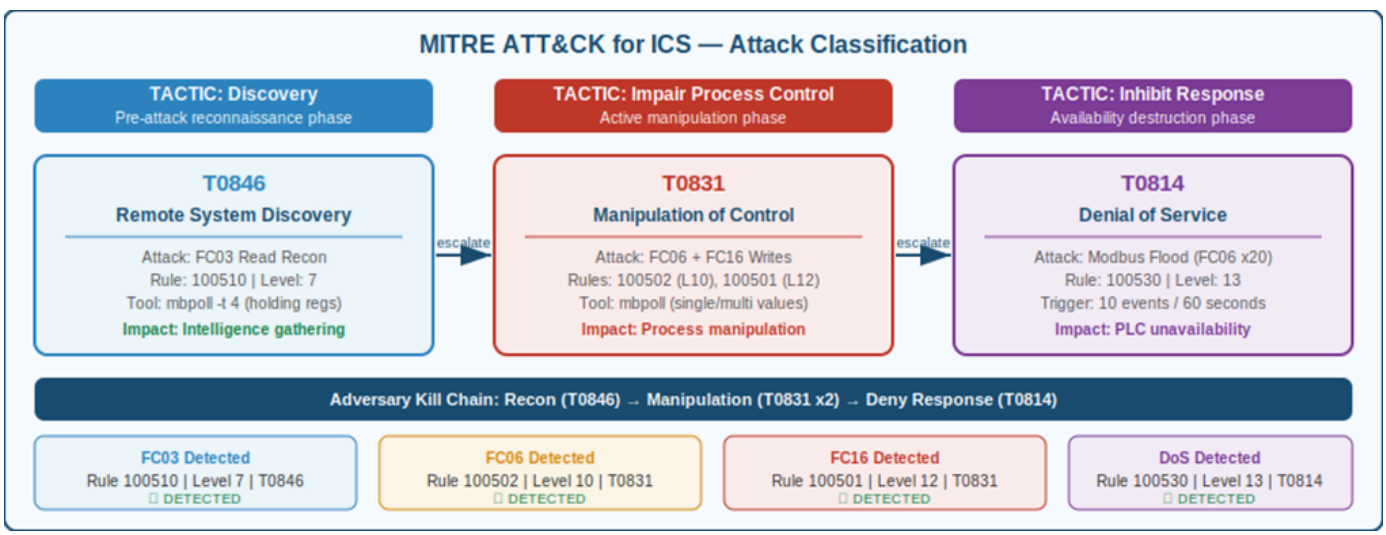
A high-frequency flood of FC06 write requests directed at the PLC. PLCs have limited processing buffers and cannot handle hundreds of simultaneous requests. When flooded, legitimate operator commands are delayed or dropped — emergency shutdown commands may not execute.





MITRE ATT&CK for ICS Mapping

All detection rules are mapped to the MITRE ATT&CK for ICS framework. Wazuh 4.5.2 does not include ICS techniques (T08xx range) in its default mitre.db database. The required techniques were manually injected using Python's built-in sqlite3 module.



Expected Detection Outcomes and System Validation

The experimental framework is expected to demonstrate that all four attack scenarios can be successfully detected and classified within the detection pipeline. Multi-register write attacks are anticipated to trigger high-severity alerts at Level 12, single register modifications at Level 10, reconnaissance activity at Level 7, and denial-of-service attacks at the highest severity

level of 13. These severity levels are calibrated to reflect the operational impact of each attack type. The integration of Snort and Wazuh is expected to provide accurate protocol-level detection, effective event correlation, and clear prioritization of security events. Overall, the experiment is designed to validate a fully functional ICS intrusion detection pipeline that integrates protocol inspection, SIEM correlation, and MITRE ATT&CK mapping to enhance security monitoring in industrial environments.

Cyber Hazard and Operability (CHAZOP) Analysis

Cyber Hazard and Operability (CHAZOP) analysis is applied in this research to systematically evaluate how cyber threats manifest as operational deviations within the NYMEGA Industrial Control System (ICS) environment. Unlike traditional vulnerability assessments that focus on identifying weaknesses in isolation, CHAZOP examines how adversarial actions translate into abnormal system behavior and process deviations. This is particularly important in ICS environments where attacks often leverage legitimate protocols such as Modbus and execute valid commands that appear normal at the network level but produce harmful physical effects .

The CHAZOP analysis was conducted by decomposing the NYMEGA architecture into critical nodes, including the PLC, HMI, Modbus communication channels, data historian, firewall infrastructure, and engineering access pathways. For each node, deviations were identified using adapted guidewords such as *More*, *Less*, *No*, *As Well As*, and *Other Than*. These deviations represent abnormal conditions such as increased command frequency, unauthorized control actions, missing telemetry, or unexpected communication patterns. Each deviation was then mapped to potential cyber causes, operational consequences, detection visibility, and mitigation strategies. At the PLC level, the analysis identified that the most critical hazard arises from unauthorized Modbus write operations. In the simulated attack scenario, an adversary successfully manipulated the water level register using Function Code 16, forcing the value far beyond its normal operating range. This deviation represents a *"More"* condition, where process values exceed expected thresholds. Although the command itself is protocol-compliant, its impact is significant, potentially leading to unsafe physical conditions. Traditional packet inspection fails to detect this behavior because the command is valid, whereas telemetry-based monitoring clearly identifies the anomaly through time-series deviations and heatmap intensity spikes.

Similarly, the manipulation of breaker states through repeated Modbus coil writes demonstrates an *"As Well As"* deviation, where unauthorized commands are injected alongside legitimate traffic. While PLC safety logic may reset these values, the repeated attempts create detectable behavioral patterns. This highlights an important observation: even when attacks do not immediately succeed, they still generate anomalous system behavior that can be identified through aggregated telemetry. At the communication layer, deviations such as increased command frequency (*"More"*) and unexpected function code usage (*"Other Than"*) indicate reconnaissance or automated attack activity. These behaviors often occur gradually to evade detection, making them difficult to identify using threshold-based intrusion detection systems. However, when analyzed over time, they produce distinct patterns that become visible in heatmap-based visualizations.

The data historian introduces another critical dimension to CHAZOP analysis. Deviations such as missing telemetry (*"Less"*) or falsified data (*"Other Than"*) directly impact system visibility and decision-making. A lack of telemetry may indicate communication disruption or stealthy attack behavior, while false data injection can mislead operators without triggering traditional alerts. These scenarios emphasize that both the presence and absence of data must be treated as potential indicators of compromise.From an architectural perspective, CHAZOP validates the effectiveness of implemented security controls such as network segmentation, reverse proxy deployment, and Just-In-Time (JIT) access. The reverse proxy eliminates direct HMI exposure, thereby removing an entire class of deviations related to unauthorized access. Similarly, JIT access reduces persistent privilege risks by limiting engineering access to time-bound sessions. However, CHAZOP also reveals that these controls do not eliminate all risks; instead, they shift the detection problem toward monitoring behavior within allowed boundaries.

A key outcome of this analysis is the identification of multi-stage attack patterns, where deviations propagate across multiple nodes. For example, an attacker may begin with increased network scanning activity, followed by unauthorized Modbus commands, ultimately resulting in abnormal process telemetry. No single monitoring layer captures the full attack, but the combination of CHAZOP mapping and telemetry analytics provides comprehensive visibility. Overall, the CHAZOP analysis demonstrates that ICS cyber risks are fundamentally behavioral in nature. Attacks rarely rely on invalid inputs; instead, they exploit normal system functionality in abnormal ways. This reinforces the need for behavior-based detection mechanisms, such as telemetry aggregation and heatmap visualization, which are central to the proposed research framework.

Node	*Deviation (Guideword)*	*Cyber Cause*	*Operational Impact*	*Detection Method*	*Mitigation*
PLC (Process Control)	More (Excess Water Level)	Modbus FC16 register write attack	Unsafe process conditions, incorrect system state	Telemetry spikes, Grafana time-series, heatmap intensity	Input validation, anomaly detection, alert thresholds
PLC (Breaker Control)	As Well As (Unauthorized Coil Writes)	Scripted Modbus FC05 commands	Intermittent control manipulation, equipment stress	Repeated state changes in telemetry, heatmap bursts	PLC logic safeguards, command monitoring
Modbus Communication	More (Increased Command Frequency)	Reconnaissance or automation scripts	Network congestion, timing disruption	Heatmap aggregation, traffic trend analysis	Rate limiting, IDS thresholds
Modbus Communication	Other Than (Unexpected Function Codes)	Unauthorized command injection	Process manipulation, integrity compromise	Protocol monitoring, anomaly detection	Protocol whitelisting, segmentation
Data Historian (InfluxDB)	Less (Missing Telemetry)	Communication disruption or stealth attack	Loss of visibility, delayed detection	Flatlined time-series, missing data patterns	Redundant telemetry collection, alerting
Data Historian	Other Than (False Data Injection)	Sensor manipulation or replay attack	Incorrect operator decisions	Cross-correlati on with network logs	Data validation, integrity checks

HMI / Reverse Proxy	As Well As (Unauthorized Access Attempt)	Direct OT exposure or credential misuse	Unauthorized control or monitoring	Access logs, Wazuh alerts	NGINX reverse proxy, authentication
Engineering Access (JIT)	More (Persistent Privilege)	Standing credentials, lack of access control	Long-term system compromise	Access logs, session monitoring	Just-In-Time access, session auditing
Firewall / Network	Other Than (Unauthorized Path)	Misconfigured rules	IT-to-OT exposure, increased attack surface	Firewall logs, SIEM correlation	Strict segmentation, rule validation
Telemetry System	No / Less (Data Loss)	Collector failure or network issue	Incomplete behavioral visibility	Missing heatmap regions	Monitoring redundancy, health checks
System-Wide	As Well As (Multi-Stage Attack Chain)	Combined attack techniques	Cascading failures across ICS layers	Correlated telemetry + SIEM + heatmaps	Defense-in-depth , behavior analytics

Fig: CHAZOP ANALYSIS TABLE

Results and Analysis

The implementation of the NYMEGA ICS cyber range, combined with the telemetry-driven monitoring framework, Snort-based protocol detection, Wazuh SIEM correlation, CHAZOP analysis, and NGINX reverse proxy deployment, produced several important observations regarding security visibility, network behavior, and operational monitoring within Industrial Control System (ICS) environments. The experimental environment successfully demonstrated how layered detection and segmentation mechanisms can improve visibility into both protocol-level attacks and long-term behavioral anomalies while maintaining compatibility with Operational Technology (OT) constraints. From an architectural perspective, the cyber range validated the effectiveness of segmented ICS network design using separate IT and OT environments connected through a pfSense firewall operating as an ICS DMZ gateway. Snort was deployed on the OT-facing interface to inspect cross-segment Modbus TCP traffic targeting the OpenPLC device hosted within the field network. This architecture successfully demonstrated how industrial traffic can be monitored at the boundary between enterprise and operational environments without requiring direct modifications to PLC systems themselves. The protocol-level detection framework provided visibility into Modbus communication behavior by inspecting the Modbus Application Protocol (MBAP) header and identifying function codes associated with reconnaissance activity, unauthorized register modification, and denial-of-service behavior. The custom Snort signatures targeting Function Codes 03, 06, and 16 successfully demonstrated how industrial protocol traffic can be classified according to operational intent rather than relying solely on traditional IT-based signatures. The use of protocol-aware detection at byte offset 7 within the Modbus payload enabled lightweight inspection of industrial communication while preserving compatibility with ICS traffic patterns and operational constraints.

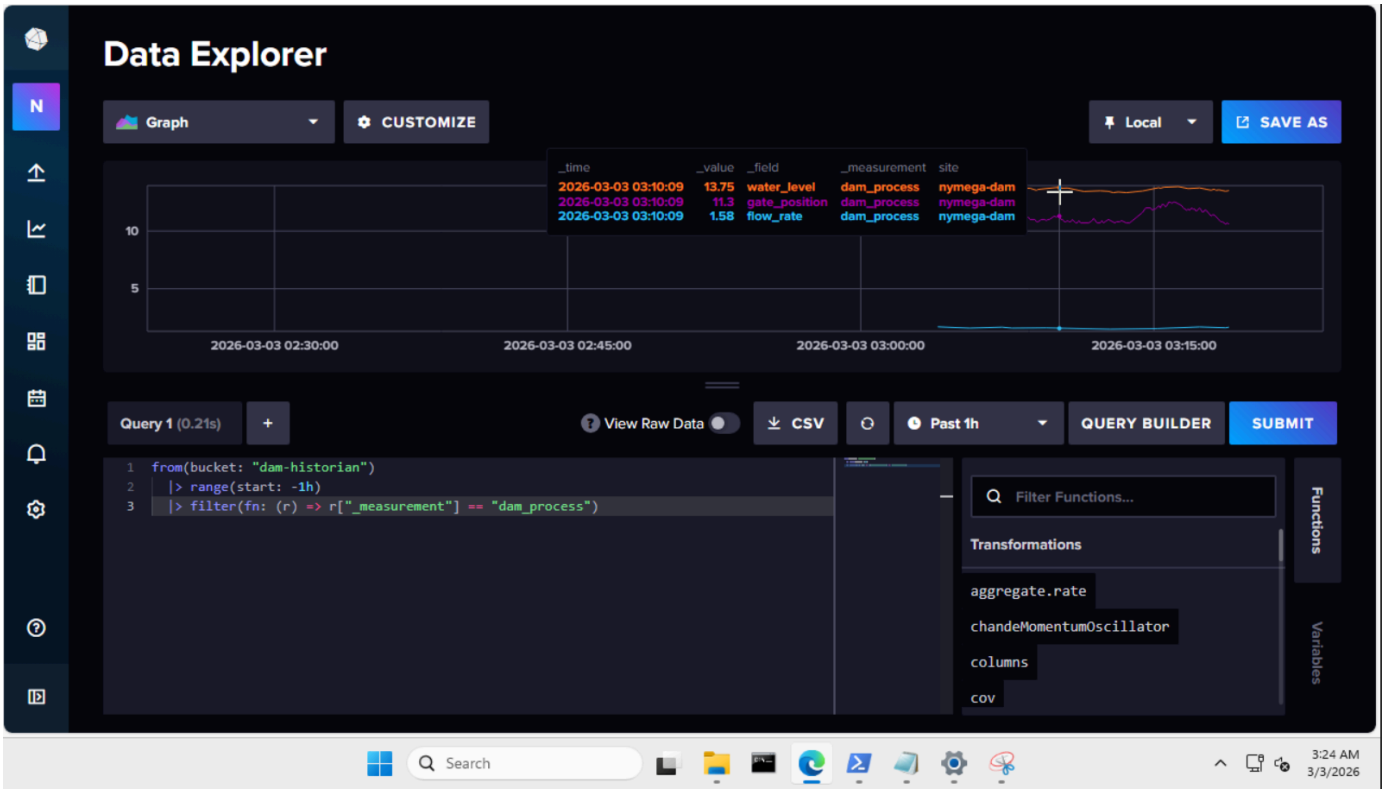
The integration of Snort with the Wazuh SIEM platform further demonstrated the importance of centralized correlation and hierarchical event classification within industrial environments. Through the use of custom decoders and parent-child rule

chaining, Modbus-related alerts were processed and classified according to operational severity and attack type. The implementation of escalating severity levels highlighted the distinction between reconnaissance behavior, targeted process manipulation, and high-frequency denial-of-service conditions. The behavioral correlation logic implemented within Rule 100530 additionally demonstrated that repeated low-level protocol events can collectively indicate larger attack patterns that may not appear malicious when viewed individually. The results also reinforced the value of MITRE ATT&CK for ICS integration within industrial monitoring environments. By mapping detected activities to standardized ICS techniques such as Manipulation of Control (T0831), Remote System Discovery (T0846), and Denial of Service (T0814), the framework provided contextualized visibility into adversarial behavior and aligned protocol-level detections with recognized industrial threat models. This significantly improved the interpretability of SIEM alerts and supported structured analysis of attack progression across the cyber range environment.

Attack Type	Rule	Level	MITRE	Status
FC03 Read Reconnaissance	100510	7 — Medium	T0846	DETECTED
FC06 Single Register Write	100502	10 — High	T0831	DETECTED
FC16 Multi-Register Write	100501	12 — Critical	T0831	DETECTED
Modbus Flood / DoS	100530	13 — Maximum	T0814	DETECTED

Attack Type	FC Code	MITRE ID	Technique Name	Tactic	Rule
FC03 Recon	FC03	T0846	Remote System Discovery	Discovery	100510 L7
FC06 Write	FC06	T0831	Manipulation of Control	Impair Process Control	100502 L10
FC16 Write	FC16	T0831	Manipulation of Control	Impair Process Control	100501 L12
DoS Flood	FC06 x10+	T0814	Denial of Service	Inhibit Response Function	100530 L13

In addition to protocol-level monitoring, the telemetry aggregation and heatmap visualization components demonstrated the importance of long-term behavioral analysis within ICS environments. Traditional packet inspection alone cannot always identify low-and-slow attacks, operational misuse, or gradual deviations in communication patterns. By aggregating Modbus telemetry over time and visualizing communication intensity across devices and function codes, the framework provided visibility into abnormal operational behavior that may otherwise blend into legitimate traffic. Heatmap-based analysis further enabled identification of communication spikes, repeated write operations, and irregular interaction patterns associated with reconnaissance or manipulation activity. CHAZOP enabled the framework to analyze how adversarial behavior impacts system operations, communication flows, and physical process stability. The analysis demonstrated that many ICS attacks rely not on invalid packets, but rather on legitimate commands executed in abnormal ways. This observation strongly supports the use of behavioral analytics and telemetry-driven monitoring within industrial environments.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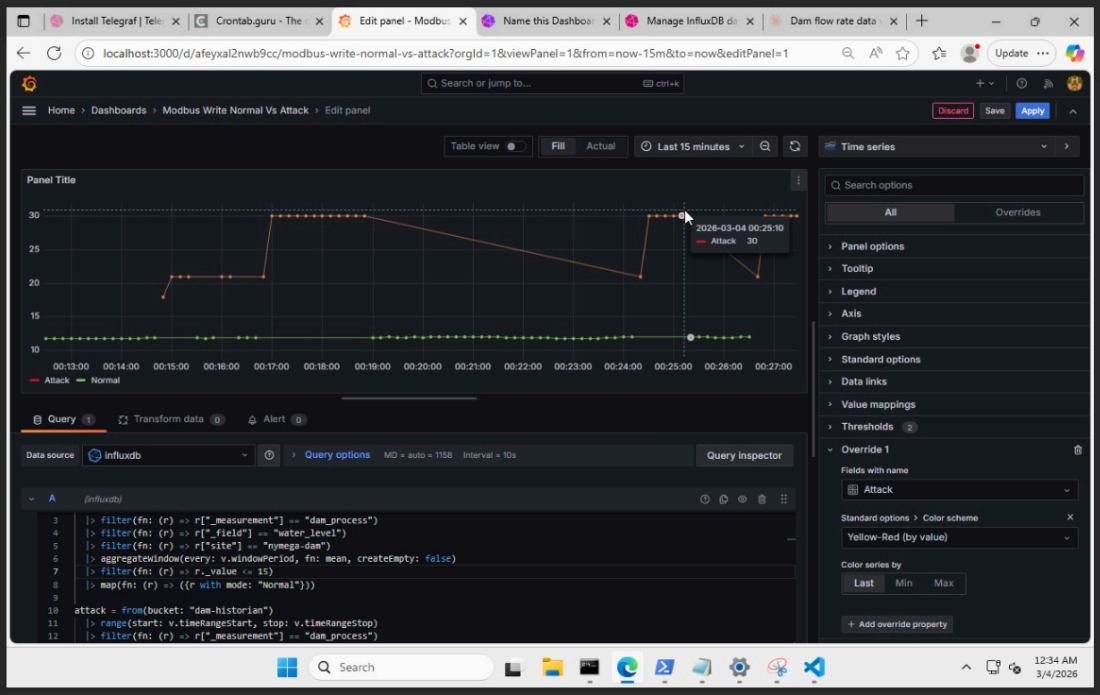
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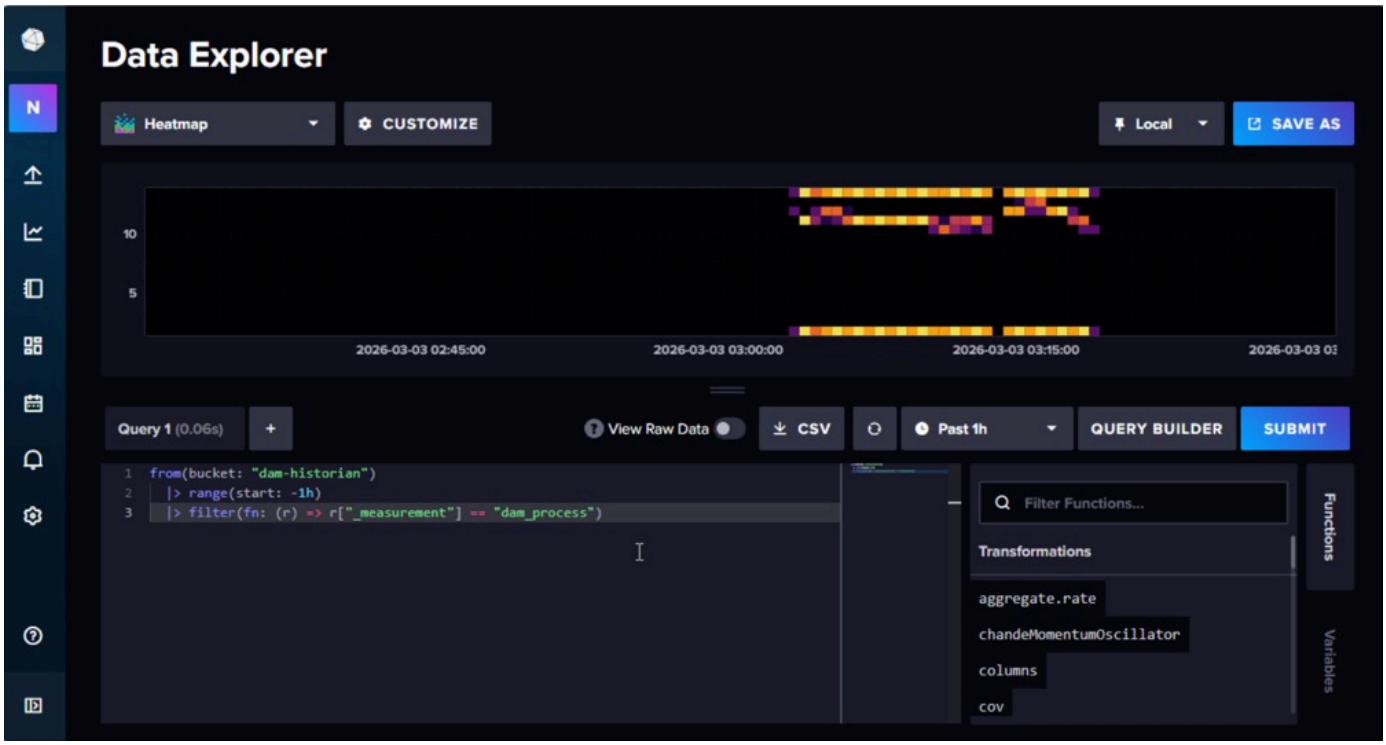
00:24:00
00:25:00
00:26:00
00:27:00
2026-03-04 00:25:10
Attack 30
Attack
Normal
Search options
All
Overrides
Panel options
Tooltip
Legend
Axis
Graph styles
Standard options
Data links
Value mappings
Thresholds 2
Override 1
Fields with name
Attack
Standard options › Color scheme
Yellow-Red (by value)
Color series by
Last
Min
Max
+ Add override property
Query 1
Transform data 0
Alert 0
Data source
influxdb
Query options
MD = auto = 1158
Interval = 10s
Query inspector
A
(influxdb)
|> filter(fn: (r) => r["_measurement"] == "dam_process")
|> filter(fn: (r) => r["_field"] == "water_level")
|> filter(fn: (r) => r["site"] == "nymega-dam")
|> aggregateWindow(every: v.windowPeriod, fn: mean, createEmpty: false)
|> filter(fn: (r) => r._value <= 15)
|> map(fn: (r) => ({r with mode: "Normal"}))
attack = from(bucket: "dam-historian")
|> range(start: v.timeRangeStart, stop: v.timeRangeStop)
|> filter(fn: (r) => r["_measurement"] == "dam_process")
Search
12:34 AM
3/4/2026



At the PLC level, CHAZOP analysis identified unauthorized Modbus write operations as one of the most critical operational hazards within the environment. Function Code 16 write operations represented “More” deviations in which process values exceeded expected operational thresholds, while repeated coil write attempts represented “As Well As” deviations involving unauthorized control actions alongside legitimate traffic. Similarly, communication-layer deviations such as excessive command frequency, unexpected function code usage, and telemetry loss highlighted the importance of monitoring both the presence and absence of operational data. These findings demonstrated that operational context is essential for accurately evaluating cyber threats within ICS environments. The deployment of the NGINX-based TCP reverse proxy further expanded the architectural security model by introducing a controlled intermediary communication layer between network segments. The implementation validated that NGINX could successfully operate as a Layer 4 reverse proxy using the stream module while maintaining compatibility with industrial TCP communication patterns. Connection validation tests confirmed that the proxy correctly accepted incoming client sessions, demonstrating successful module configuration, listener binding, and process execution.

From a network security perspective, the reverse proxy reinforced defense-in-depth principles by abstracting direct communication between clients and backend control systems. Rather than exposing OT assets directly to upstream networks, communication was funneled through a centralized intermediary node capable of enforcing structured access paths and supporting additional monitoring visibility. This reduced the exposure of critical ICS devices and limited opportunities for unauthorized lateral movement between network zones. The proxy deployment also highlighted an important distinction between network reachability and application-layer service availability. While ICMP connectivity tests confirmed that backend systems were reachable, TCP connection attempts demonstrated that backend services were not always actively listening or accepting sessions on the expected ports. This observation emphasized a common characteristic of industrial environments, where strict communication policies, segmented architectures, and restricted service exposure may intentionally prevent direct protocol access even when network connectivity exists. Importantly, these observations did not indicate failure of the reverse proxy itself, but rather reflected environmental constraints and backend service configurations.

The experiments additionally demonstrated the differences between HTTP-based reverse proxy validation and TCP-based industrial proxying. Unlike traditional web services that can be tested using browser interactions, industrial TCP proxying requires protocol-aware validation techniques that operate at the transport layer. Successful acceptance of TCP sessions by NGINX confirmed proper transport-layer forwarding functionality even in situations where application-layer responses were unavailable. When viewed together, the combined implementation of Snort IDS, Wazuh SIEM correlation, telemetry analytics,

CHAZOP operational analysis, and NGINX reverse proxying demonstrated the effectiveness of layered monitoring and segmentation strategies within industrial environments. The results show that no single monitoring mechanism provides complete visibility into ICS threats independently. Packet inspection identifies protocol activity, telemetry analytics reveals long-term behavioral deviations, SIEM correlation links related events across systems, CHAZOP maps cyber activity to operational impact, and reverse proxy segmentation reduces exposure while centralizing communication visibility. Overall, the research demonstrates that behavior-based monitoring, layered detection architectures, and operationally aware security analysis significantly improve visibility into stealthy cyber threats within ICS environments while maintaining the availability and stability requirements essential to operational technology systems.

Future Work

While the current implementation successfully demonstrates the feasibility of using NGINX as a TCP reverse proxy within an ICS environment, several opportunities exist for extending and enhancing this work to improve both functionality and security. One key area for future development is the integration of secure communication mechanisms at the proxy layer. Industrial protocols such as Modbus are inherently insecure, lacking encryption and authentication. By incorporating Transport Layer Security (TLS) at the NGINX proxy, it would be possible to encrypt communication between clients and the proxy, thereby protecting sensitive control data from interception. This approach would allow legacy ICS protocols to benefit from modern security controls without requiring modifications to the underlying devices. Another important extension involves implementing authentication and access control mechanisms at the proxy level. Currently, the proxy forwards traffic without verifying client identity. Introducing authentication methods such as certificate-based access control or token-based validation would enable fine-grained control over which users or systems are permitted to interact with ICS components. This is particularly relevant in environments where remote access to control systems must be tightly regulated.

Further work can also explore the integration of NGINX with centralized monitoring and logging systems. By exporting connection logs and session metadata to platforms such as Wazuh or other SIEM solutions, the proxy can contribute to a more comprehensive security monitoring framework. This would enable correlation between network-level events, system logs, and telemetry data, improving the detection of anomalous or malicious behavior. Another potential enhancement is the incorporation of rate limiting and anomaly detection at the proxy layer. Given that many ICS attacks rely on high-frequency command execution or abnormal communication patterns, implementing traffic shaping and threshold-based controls within NGINX could help mitigate denial-of-service attacks and unauthorized command bursts. This would add a proactive defense capability in addition to the existing detection mechanisms. Future research may also investigate the use of NGINX as part of a more advanced ICS security architecture, such as a fully implemented DMZ proxy model. In this scenario, the proxy would act as the sole interface between IT and OT networks, enforcing strict unidirectional communication policies and preventing direct access to critical control systems. This aligns with the reverse proxy architecture described in the NYMEGA environment and supports secure remote monitoring of ICS processes.

Additionally, performance evaluation under realistic industrial workloads represents an important area for further study. While the current implementation confirms functional correctness, future experiments could measure latency, throughput, and resource utilization when handling high volumes of industrial traffic. This would provide insights into the scalability of the proxy and its suitability for deployment in large-scale ICS environments. Finally, future work can focus on combining the proxy layer with advanced analytics techniques, such as machine learning–based anomaly detection. By integrating proxy-level telemetry with historian data and behavioral models, it would be possible to develop a unified detection framework capable of identifying both network-level and process-level anomalies. This would significantly enhance situational awareness and strengthen the overall security posture of industrial control systems.

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